Xiaowen Zhang

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Research Interests

- **Cosmological simulations:** large-scale structure, N-body and hydrodynamic simulations, Lyman-α forest
- **Machine learning for science:** deep generative models (GANs, diffusion, flow matching), graph neural networks, surrogate modeling and emulation, foundation models and their interpretability
- **Computation:** HPC and large-scale GPU training, LLM multi-agent systems for research automation

Education

- **Carnegie Mellon University** — *Sep. 2021 - Present*
  Ph.D. in Physics, expected 2027 (Advisors: Prof. Tiziana Di Matteo and Prof. Rupert Croft) — Pittsburgh, USA
- **ShanghaiTech University** — *Sep. 2016 - Jun. 2020*
  B.S. in Physics — Shanghai, China
- **University of California, Berkeley** — *Aug. 2018 - May 2019*
  Visiting Student, Physics Department — Berkeley, USA

Research Experience

- **McWilliams Center for Cosmology & Astrophysics** — *Sep. 2021 - Present*
  Graduate Research Assistant — Carnegie Mellon University
  - Developed a time-conditioned **style-based GAN** that super-resolves N-body simulations by **512**$\times$ in particle count, preserving matter power spectra and halo mass functions and producing time-consistent particle evolution across cosmic time.
  - Built a **deterministic emulator** conditioned on high-resolution initial conditions, converting stochastic super-resolution output into the specific target realization; benchmarked against full N-body runs on halo statistics, merger trees, and Fourier-space cross-correlation.
  - Designed a two-stage model emulating high-resolution hydrodynamic fields (gas, stars) from low-resolution runs at $z=3$, achieving **sub-percent** errors on overdensity, temperature, velocity, and optical-depth fields at $\sim$**450**$\times$ the speed of SPH at equal resolution, enabling high-fidelity Lyman-α sightline prediction.
  - Developing a **foundation model for cosmological simulations**: a shared embedding space that ingests heterogeneous simulation outputs across codes, resolutions, and modalities, designed to be invariant to initial-condition realizations while remaining sensitive to cosmological parameters. Current Set Transformer models over variable-size halo catalogs raise matched Ω_m recovery to $R^2 \approx 0.85$. First author, manuscript in preparation.
  - Analyzing the physics encoded in the embeddings of **AION-1**, an omnimodal foundation model for astronomical observations, in collaboration with Prof. Barnabas Poczos (CMU Machine Learning Department).
  - Built **SimAgents**, a multi-agent LLM system (multi-agent debate and retrieval-augmented generation) in which agents read literature and simulation manuals, extract configuration parameters, auto-generate MP-GADGET configs, and run first-look analyses, reaching $>$**98% end-to-end parameter-extraction accuracy**.
  - Contributing to **agentic simulation calibration** and to a curated model zoo for astrophysical simulation-based inference, both part of the group's AI Virtual Universe programme.
  - Training a message-passing **graph neural network** on MP-GADGET outputs to learn fluid evolution; comparing accuracy and stability against traditional numerical solvers.
  - Ran multi-node, multi-GPU training campaigns on national HPC systems, building the data pipelines and training infrastructure.
- **Institute of Mathematical Sciences** — *Aug. 2020 - Aug. 2021*
  Research Assistant — ShanghaiTech University
  - Applied **topological data analysis** to characterize manifold curvature and connectivity in high-dimensional cosmological datasets.
  - Investigated **tropical geometry** representations of neural network decision boundaries, connecting algebraic structure to model behavior.

Publications

[1] **Zhang, X.**, Lachance, P., Ni, Y., Li, Y., Croft, R., Di Matteo, T., Bird, S., Feng, Y., et al. (2024). *AI-assisted super-resolution cosmological simulations III: time evolution*. *MNRAS*, **528**, 281–293.

[2] **Zhang, X.**, Lachance, P., Dasgupta, A., Croft, R., Di Matteo, T., Ni, Y., Bird, S., & Li, Y. (2025). *AI-assisted super-resolution cosmological simulations IV: An emulator for deterministic realizations*. *The Open Journal of Astrophysics*, **8**, E13.

[3] **Zhang, X.**, Bi, Z., LaChance, P., Wang, X., Di Matteo, T., & Croft, R. (2025). *SimAgents: Bridging Literature and the Universe Via A Multi-Agent Large Language Model System*. *Proc. IJCNLP-AACL 2025: System Demonstrations*, 55–66. Association for Computational Linguistics.

[4] Hafezianzadeh, F.*, **Zhang, X.**, Ni, Y., Croft, R. A. C., Di Matteo, T., Qezlou, M., & Bird, S. (2025). *An AI super-resolution field emulator for cosmological hydrodynamics: the Lyman-α forest*. arXiv:2507.16189.

[5] Zhou, Y., Di Matteo, T., Bird, S., Croft, R. A. C., Ni, Y., Yang, Y., Chen, N., Lachance, P., **Zhang, X.**, & Hafezianzadeh, F.*. (2025). *The ASTRID Simulation at $z = 0$: from Massive Black Holes to Large-scale Structure*. arXiv:2510.13976.

**Denotes mentored student.*

In preparation

[6] **Zhang, X.**, et al. *A shared embedding space for astrophysical simulations*. Manuscript in preparation (first author).

[7] Physics interpretability of AION-1 embeddings. In progress, with the CMU Machine Learning Department.

Talks & Presentations

- **2nd Annual Cosmic Horizons Conference (CosmicAI)** *Jul. 2026*
  Talk: *AI Accelerators for Cosmological Simulations: Super-Resolution and Emulation* Charlottesville, VA
- **COSMO-25: 28th Int. Conference on Particle Physics & Cosmology** *Oct. 2025*
  Talk: *Accelerating cosmological simulations with super resolution & Application of LLM in running simulations* Carnegie Mellon University
- **Machine Learning in Astronomical Surveys** *Nov. 2023*
  Poster: *AI-assisted Super-Resolution Cosmological Simulations IV: An Emulator for Deterministic Realizations* Flatiron Institute
- **Cosmic Connections: A ML $\times$ Astrophysics Symposium** *May 2023*
  Poster: *AI-assisted Super-Resolution Cosmological Simulations III: Time Evolution* Flatiron Institute

Grants & Awards

- **ACCESS HPC Allocation** *2025-*
  Learning Cosmological Simulations with Graph Neural Networks **PI**, PHY250036
- **ACCESS HPC Allocation** *2024-2025*
  An Emulator for Transforming Random Super-Resolution Realizations to Deterministic High-Resolution Outputs PHY240032
- **Block Center Seed Fund**
  AI Acceleration of Scientific Research (team award, supporting SimAgents) Carnegie Mellon University
- **Scholarship of Academic Excellence**
  ShanghaiTech University 2017
- **Dean's Scholarship**
  ShanghaiTech University 2016

Teaching & Mentoring

- **Research Mentor**
  - **Fatemeh Hafezianzadeh** (graduate), *Super-resolution of Lyman-α Forest Simulations*, 2024-present — first author of publication [4].
  - **Ankita Dasgupta** (undergraduate), *An Emulator for Deterministic Super-Resolution Output*, 2023-2024 — co-author of publication [2].
  - **Aidan Catalano & Matthew Pan** (undergraduates), *Learning Cosmological Simulations with GNNs*, 2025-present.
- **Graduate Teaching Assistant, Carnegie Mellon University**
  - *33-152 Matter & Interactions II*; *33-142/122 Introductory Physics II* — led weekly recitations and office hours, graded coursework.
  - *33-231 Physical Analysis* — led weekly office hours, graded coursework.
  - *33-341 Thermal Physics I* — grader.
  - **Course design:** contributed to course design for *[course number and name] — [what you designed, with whom, when]*.

Open Source Software

- **map2map** : Python/**PyTorch** style-based GAN framework for super-resolving N-body simulations to **512**$\times$ higher mass resolution.
- **SREmulator** : Neural models that transform stochastic super-resolution outputs into deterministic realizations matching specific initial conditions.

Technical Skills

- **Languages & Tools:** Python (**PyTorch**, JAX, TensorFlow, NumPy, SciPy, Pandas), MATLAB, C++ (basic), LaTeX, Git
- **Machine Learning:** Deep generative models (GANs, diffusion, flow matching), graph neural networks, convex optimization, LLM multi-agent systems and RAG
- **Computing & Domain:** CUDA, multi-node multi-GPU training, HPC, large-scale data pipelines; GADGET, MP-GADGET
- **Graduate Coursework:** 10-701 Machine Learning (A); 10-725 Convex Optimization (A–), CMU Machine Learning Department

Languages

- Mandarin Chinese (native); English (fluent — research, teaching, and conference medium since 2021).

Doctoral Committee & References

- **Advisors**
  - **Prof. Tiziana Di Matteo** — Professor of Physics; Director, McWilliams Center for Cosmology & Astrophysics, Carnegie Mellon University tizianad@andrew.cmu.edu
  - **Prof. Rupert A. C. Croft** — Professor of Physics, Carnegie Mellon University rcroft@cmu.edu
- **External committee members**
  - **Prof. Barnabas Poczos** — Machine Learning Department, Carnegie Mellon University.
  - **Dr. Francisco Villaescusa-Navarro** — Research Scientist, Cosmology, Center for Computational Astrophysics, Flatiron Institute.
- Letters of reference available upon request.